

Analysis

Anonymous

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TPOT with full graph search space versus ensemble

2000 total evals

Version 1

Stats Testing

```
## # A tibble: 26 x 9
##   task_id .y. group1      group2    n1    n2 statistic      p p.signif
##   <int> <chr> <chr>      <chr> <int> <int>    <dbl>    <dbl> <chr>
## 1   2073 auroc ensemble_random_~ indiv~    16    16    119 6.29e-3 **
## 2  146818 auroc ensemble_random_~ indiv~    19    19    148. 6.48e-3 **
## 3  146820 auroc ensemble_random_~ indiv~    21    21    182 2.18e-2 *
## 4  168350 auroc ensemble_random_~ indiv~    21    21    227 6.68e-6 ****
## 5  168757 auroc ensemble_random_~ indiv~    19    19    163 6.6 e-3 **
## 6  168784 auroc ensemble_random_~ indiv~    19    19    177 3.36e-4 ***
## 7  168911 auroc ensemble_random_~ indiv~    19    19    161 6.18e-3 **
## 8  190137 auroc ensemble_random_~ indiv~    21    21    225 1.34e-5 ****
## 9  190146 auroc ensemble_random_~ indiv~    21    21    148 2.72e-1 ns
## 10 190411 auroc ensemble_random_~ indiv~    21    21    193 5.54e-3 **
## 11 359955 auroc ensemble_random_~ indiv~    21    21    170 5.95e-2 ns
## 12 359956 auroc ensemble_random_~ indiv~    18    18    160 4.2 e-4 ***
## 13 359958 auroc ensemble_random_~ indiv~    21    21    215 1.61e-4 ***
## 14 359959 auroc ensemble_random_~ indiv~    21    21    225 1.34e-5 ****
## 15 359960 auroc ensemble_random_~ indiv~    20    20     50 2.49e-2 *
## 16 359962 auroc ensemble_random_~ indiv~    19    19    147 3.61e-2 *
## 17 359963 auroc ensemble_random_~ indiv~    21    21    228 4.77e-6 ****
## 18 359964 auroc ensemble_random_~ indiv~    18    18    144 8.96e-3 **
## 19 359965 auroc ensemble_random_~ indiv~    21    21    118 1.63e-1 ns
## 20 359968 auroc ensemble_random_~ indiv~    21    21    149 2.57e-1 ns
## 21 359969 auroc ensemble_random_~ indiv~    21    21    231 9.54e-7 ****
## 22 359970 auroc ensemble_random_~ indiv~    21    21    231 9.54e-7 ****
## 23 359971 auroc ensemble_random_~ indiv~    21    21    212 2.93e-4 ***
## 24 359972 auroc ensemble_random_~ indiv~    21    21    216 1.31e-4 ***
## 25 359974 auroc ensemble_random_~ indiv~    18    18    160 4.2 e-4 ***
## 26 359975 auroc ensemble_random_~ indiv~    21    21    122 8.38e-1 ns
```

Medians

```
## # A tibble: 26 x 3
##   task_id individual_tpot ensemble_random_100
##   <int>          <dbl>          <dbl>
## 1   2073          0.878          0.887
## 2  146818          0.936          0.937
## 3  146820          0.995          0.997
## 4  168350          0.965          0.967
## 5  168757          0.790          0.801
## 6  168784          0.962          0.969
## 7  168911          0.868          0.872
## 8  190137          0.919          0.931
## 9  190146          0.967          0.968
## 10 190411          0.921          0.923
## 11 359955          0.743          0.744
## 12 359956          0.932          0.938
## 13 359958          0.942          0.948
## 14 359959          0.742          0.755
## 15 359960          1          1
## 16 359962          0.837          0.845
## 17 359963          0.995          0.996
## 18 359964          0.993          0.994
## 19 359965          1.00          1.00
```

##	20	359968	0.929	0.926
##	21	359969	0.845	0.854
##	22	359970	0.914	0.930
##	23	359971	0.997	0.998
##	24	359972	0.991	0.992
##	25	359974	0.864	0.889
##	26	359975	0.994	0.995